M6 · Complement deficiencies, regulator defects and hereditary angioedema

Innate immunity lecture pack · EAACI/UEMS Knowledge Exam preparation

After this module you can

- Link each complement deficiency to its typical phenotype and explain why the phenotype follows from the lost function.
- Read a complement and C1 inhibitor (C1-INH) laboratory profile and separate HAE-1, HAE-2, HAE with normal C1-INH and acquired C1-INH deficiency.
- Explain the bradykinin pathway of hereditary angioedema (HAE) and why antihistamines, glucocorticoids and adrenaline do not work.
- Choose on-demand treatment, short-term prophylaxis and long-term prophylaxis according to the 2017 WAO/EAACI guideline.
- Recognise PNH and aHUS as diseases of defective complement regulators.

AAE-C1-INH	acquired angioedema due to C1-INH deficiency
ACE / ACEI-AE	angiotensin-converting enzyme / ACE inhibitor-induced angioedema
aHUS	atypical haemolytic uraemic syndrome
C1-INH	C1 inhibitor
C3NeF	C3 nephritic factor
CH50	total classical pathway haemolytic activity
CSU	chronic spontaneous urticaria
DAF	decay-accelerating factor (CD55)
FFP / SDP	fresh frozen plasma / solvent detergent-treated plasma
FXII	coagulation factor XII
GPI	glycosylphosphatidylinositol
HAE	hereditary angioedema (HAE-1, HAE-2, HAE nC1-INH)
HMWK	high-molecular-weight kininogen
LTP / STP	long-term prophylaxis / short-term (pre-procedural) prophylaxis
MAC	membrane attack complex
MASP	MBL-associated serine protease
MBL	mannose-binding lectin
MCP	membrane cofactor protein (CD46)
MGUS	monoclonal gammopathy of undetermined significance
pdC1-INH / rhC1-INH	plasma-derived / recombinant human C1-INH
PNH	paroxysmal nocturnal haemoglobinuria
SLE	systemic lupus erythematosus

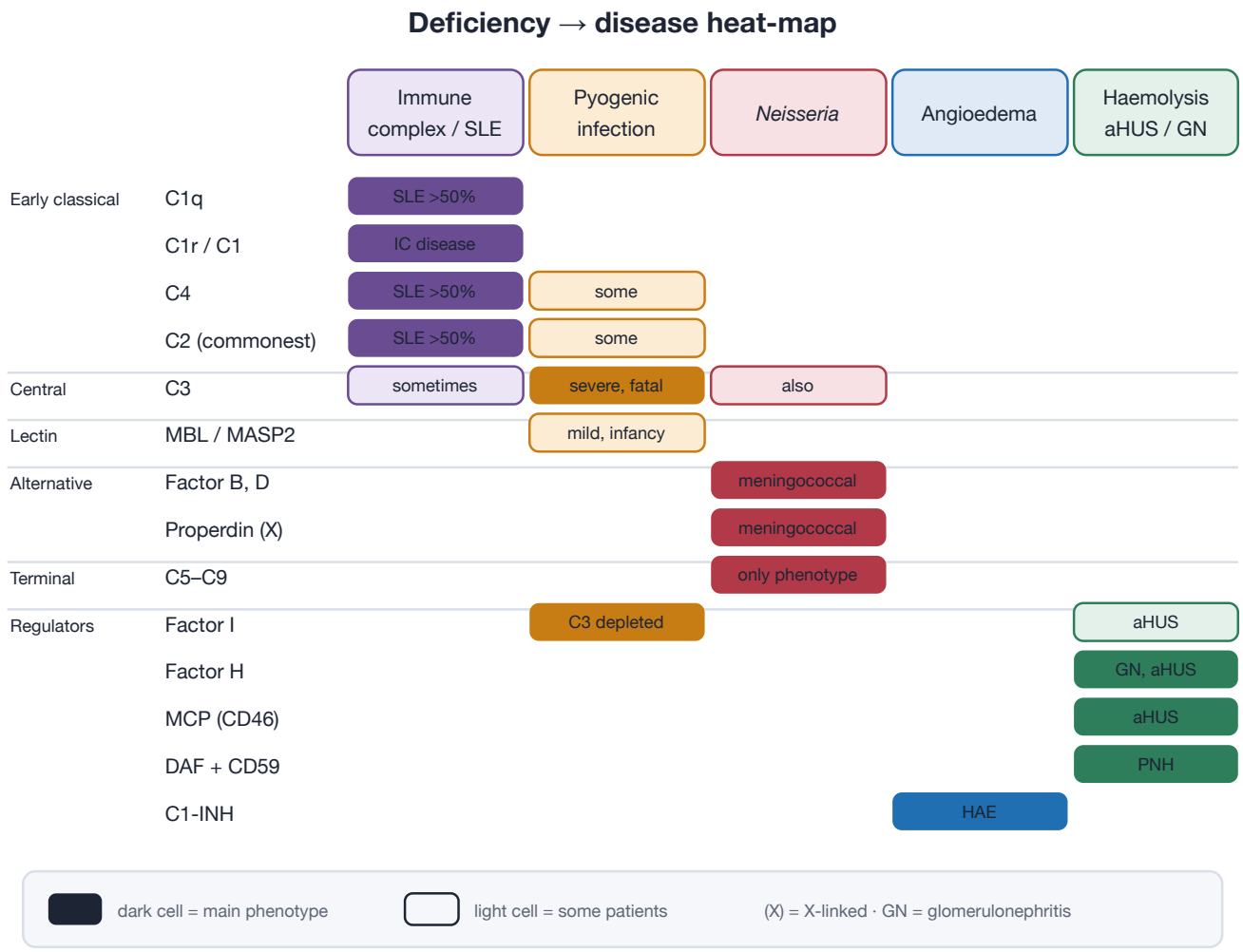
1. The rule: the lost function predicts the disease

In M5 you learned three jobs of complement: clear immune complexes, opsonize bacteria with C3b, and lyse some bacteria with the membrane attack complex (MAC). You also learned that regulators protect host cells. Each deficiency removes one job. The disease shows which job is missing.

1. Loss of an early classical component (C1q, C1r, C4, C2) → poor clearance of immune complexes and apoptotic cells → SLE-like immune-complex disease. Janeway p.532
2. Loss of C3 (or C3 depletion) → poor opsonization → severe pyogenic infections. Abbas p.315
3. Loss of a terminal component (C5–C9), properdin or factor D → *Neisseria* infections. Janeway p.532
4. Loss of a regulator → uncontrolled activation → C3 consumption, aHUS, PNH or (for C1-INH) bradykinin angioedema. Abbas p.315 HAE-GL p.6

> **Analogy**
>
> Think of complement as a factory line with three output gates: "waste disposal" (immune complexes), "tagging" (C3b) and "shredder" (MAC). Break the waste-disposal gate and the factory clogs (SLE). Break the tagging station and nothing is labelled for the phagocyte (pyogenic infection). Break only the shredder and only thin-walled packages survive (*Neisseria*). Break the brakes and the line runs out of control.

2. Deficiency → phenotype map



D18. Pathway heat-map. Early classical components → immune-complex (SLE-like) disease; C3 → pyogenic infection; alternative and terminal components → *Neisseria*; regulators → C3 consumption, aHUS, PNH and HAE. Abbas p.315 Janeway p.532–533

2.1 Early classical components: C1q, C1r, C4, C2 → immune-complex disease

Genetic deficiencies of C1q, C1r, C4, C2 and C3 occur in humans. Abbas p.315 C2 deficiency is the most common human complement deficiency. Abbas p.315 More than 50% of patients with C1q, C2 or C4 deficiency develop SLE. Abbas p.315

Why? Early classical components mainly process immune complexes and clear apoptotic cells. Their loss causes immune-complex disease. Janeway p.532 Uncleared immune complexes deposit in vessel walls and tissues and cause inflammation. Abbas p.315 Apoptotic bodies that are not cleared are a likely source of the nuclear autoantigens of lupus. Abbas p.315 Some patients with C2 or C4 deficiency also have more infections; others have no symptoms. Abbas p.315

> **Exam pearl**
>
> Young patient with SLE-like disease + early classical deficiency → think C1q, C4 or C2. C2 is the most common deficiency overall. Abbas p.315

Separate data for C1s deficiency are not given. Not covered in provided sources — needs expert input.

2.2 C3 and the lectin pathway → pyogenic infection

C3 deficiency causes frequent, serious pyogenic infections that can be fatal. This shows the central role of C3 in opsonization and phagocytosis. Abbas p.315 Defects in C3 or in C3 activation increase susceptibility to pyogenic bacteria, including *Streptococcus pneumoniae*. Janeway p.532 Janeway also lists *Neisseria* infection and, sometimes, immune-complex disease in C3 deficiency. Janeway p.532

Mannose-binding lectin (MBL) deficiency is relatively common (about 5% of the population). Janeway p.533 It may cause a mild immunodeficiency with more bacterial infections in early childhood. MASP2 defects give a similar phenotype. Janeway p.533

2.3 Alternative and terminal components → *Neisseria*

Deficiencies of alternative pathway components increase susceptibility to meningococcal infections. Abbas p.315 Factor B and factor D deficiencies are rare. X-linked recessive properdin deficiency is more common. Abbas p.315 Properdin-deficient patients are particularly susceptible to *Neisseria meningitidis*. Janeway p.59

In C5, C6, C7, C8 or C9 deficiency, the only consistent clinical problem is disseminated *Neisseria* infection. MAC lysis is important against these thin-walled bacteria. Abbas p.315 Defects in factor D and properdin give a similar *Neisseria* susceptibility. Janeway p.532 In a Japanese population study, the yearly risk of meningococcal infection was about 1 in 200 with an inherited MAC protein deficiency, versus 1 in 2,000,000 in normal people (10,000-fold). Janeway p.533

> **Clinical correlation**
>
> Recurrent or disseminated *Neisseria* infection → screen for terminal component or alternative pathway deficiency. A male patient with an affected maternal relative suggests X-linked properdin deficiency. Abbas p.315

2.4 Regulator deficiencies → uncontrolled activation

Factor I. Without factor I, fluid-phase C3 convertase forms without control and depletes plasma C3. Abbas p.315 The result is a de facto C3 deficiency. Janeway p.533 The patient therefore has more pyogenic infections, like a patient with C3 deficiency. Abbas p.315

Factor H. Factor H blocks the binding of Bb to C3b. It regulates the alternative pathway but not the classical pathway. Abbas p.311 Factor H deficiency is rare. It causes excess alternative pathway activation, C3 consumption and glomerulonephritis. Abbas p.315 The autoantibody C3 nephritic factor (C3NeF) has a

similar effect: it stabilizes C3bBb against factor H and consumes C3. Patients often have glomerulonephritis. Abbas p.315 The term "C3 glomerulopathy" is not used in the sources. Not covered in provided sources — needs expert input.

MCP, factor H, factor I → aHUS. See section 5. Abbas p.315 Janeway p.70

DAF and CD59 → PNH. See section 5. Abbas p.315

C1-INH → HAE. See section 4. Abbas p.311

Lost protein	Function lost (M5)	Main phenotype	Source
C1q, C1r, C4, C2	Clearance of immune complexes and apoptotic cells	SLE-like disease (>50% for C1q, C2, C4); some infections	Abbas p.315; Janeway p.532
C3	Opsonization (C3b)	Severe pyogenic infections	Abbas p.315; Janeway p.532
MBL / MASP2	Lectin pathway start	Mild; bacterial infections in early childhood	Janeway p.533
Factor B, D, properdin (X-linked)	Alternative pathway amplification	Meningococcal infection	Abbas p.315; Janeway p.532
C5–C9	MAC lysis	*Neisseria* only	Abbas p.315; Janeway p.532
Factor I	Cleavage of C3b	Secondary C3 depletion → pyogenic infections	Abbas p.315
Factor H	Alternative pathway brake	C3 consumption, glomerulonephritis; aHUS	Abbas p.315
DAF + CD59 (GPI anchor, PIG-A)	Protection of host cells	PNH	Abbas p.315
C1-INH	Brake on C1r/C1s, MASP, kallikrein, FXIIa	HAE (bradykinin)	HAE-GL p.6

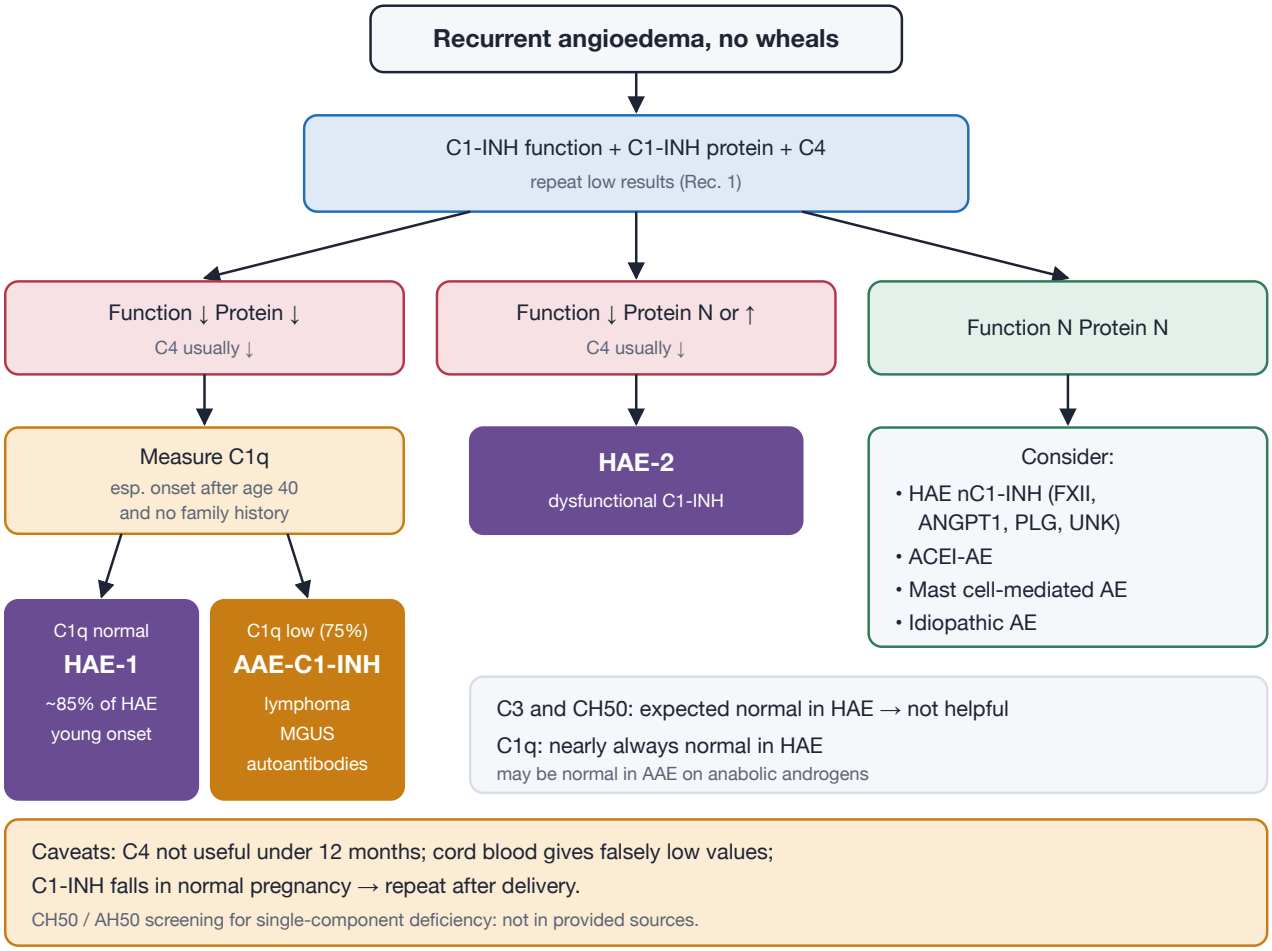
3. Laboratory work-up

The sources give a clear algorithm for suspected C1-INH deficiency. They do not explain how to use CH50 or AH50 to screen for single component deficiencies. Not covered in provided sources — needs expert input.

1. Suspect HAE in recurrent angioedema without wheals, especially with family history, onset in childhood, abdominal attacks or upper airway swelling. HAE-GL p.6
2. Measure C1-INH function, C1-INH protein and C4 in every suspected patient. HAE-GL p.6
3. Repeat any abnormally low result to confirm the diagnosis. HAE-GL p.6

4. Do not rely on C3 and CH50. They are expected to be normal in HAE and are usually not helpful. HAE-GL p.6–7

5. Measure C1q when acquired deficiency is possible, especially new-onset angioedema after age 40 with no family history. HAE-GL p.7



D19. Laboratory decision tree for suspected C1-INH deficiency. HAE-GL p.5–7 HAE-GL p.7 HAE-GL p.11–12

Condition	C1-INH function	C1-INH protein	C4	C1q	Clues
HAE-1 (~85%)	↓	↓	usually ↓	normal	Family history (absent in up to 25%); onset in childhood
HAE-2	↓	N or ↑	usually ↓	normal	As HAE-1
AAE-C1-INH	↓	↓	↓	↓ in 75%	Later onset; lymphoma, MGUS, B symptoms; autoantibodies to C1-INH
HAE nC1-INH	N	N	not stated in sources		Family history; FXII, ANGPT1, PLG or unknown mutation

Sources for the table: HAE-1 has low antigenic and functional C1-INH. HAE-GL p.5 HAE-2 has normal or high antigenic but low functional C1-INH. HAE-GL p.5 HAE-1 is about 85% of patients. C4 is usually low, but its sensitivity and specificity are limited. HAE-GL p.6 A family history may be absent in up to 25% of patients. HAE-GL p.6 AAE-C1-INH has the same basic profile as HAE-1. It differs by later onset, underlying lymphoma or MGUS, occasional B symptoms and often low C1q. HAE-GL p.7 C1q is nearly always normal in HAE and low in 75% of AAE-C1-INH. HAE-GL p.7 Many AAE-C1-INH patients have autoantibodies that inactivate C1-INH. HAE-GL p.7

SERPING1 sequencing can support the diagnosis, including prenatal diagnosis. Biochemical C1-INH testing is effective and cheaper, and sequencing can miss some mutations. HAE-GL p.7

> **Exam pearl**
>
> Low C4 is a screening clue, not proof. The diagnosis rests on C1-INH protein and function, repeated. HAE-GL p.6 Low C1q in a 60-year-old with new angioedema points to acquired deficiency. HAE-GL p.7

4. Hereditary angioedema in depth

> **Check the current guideline**
>
> All treatment recommendations below follow the WAO/EAACI guideline, **2017 revision** (Allergy 2018). A 2021 revision exists and is not in the provided sources. Check every drug, licence, age limit and first-line choice against the current guideline before the exam. Volatile statements are marked with a dashed citation.

4.1 Definition and types

Angioedema can be mediated by bradykinin and/or by mast cell mediators such as histamine. HAE-GL p.5 The guideline names these hereditary and acquired forms:

- **HAE-1**: C1-INH deficiency; low antigenic and low functional C1-INH. HAE-GL p.5
- **HAE-2**: C1-INH dysfunction; normal or high antigenic but low functional C1-INH. HAE-GL p.5
- **HAE nC1-INH**: HAE with normal C1-INH — HAE-FXII, HAE-ANGPT1, HAE-PLG and HAE-UNK. HAE-GL p.6
- **AAE-C1-INH**: angioedema due to acquired C1-INH deficiency. HAE-GL p.5
- **ACEI-AE**: acquired, bradykinin-mediated angioedema with normal C1-INH. HAE-GL p.6

HAE nC1-INH is very rare and clinically resembles HAE-1/2. Bradykinin seems to play a major role mainly in patients with an FXII mutation. HAE-GL p.6 In most patients with HAE nC1-INH, no gene mutation is found. HAE-GL p.6 The 2017 guideline recommendations apply only to HAE-1/2. HAE-GL p.6

> **Exam pearl — "type III"**
>
> Older texts and the EAACI sample question call HAE with normal C1-INH "type III". The 2017 guideline uses "HAE nC1-INH". Typical vignette: recurrent swelling, affected relative, normal C1-INH level and function. HAE-GL p.6

4.2 Genetics

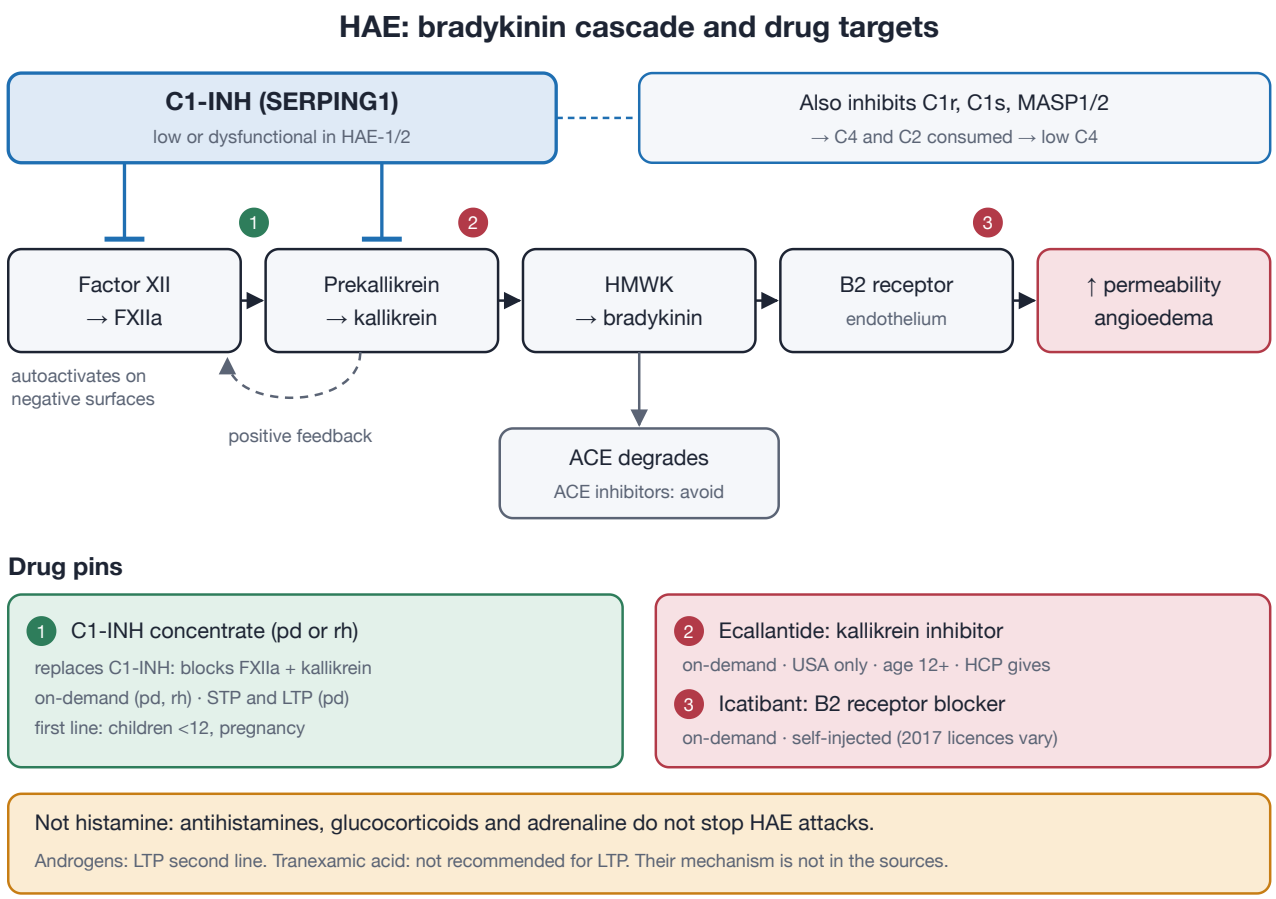
HAE-1/2 is a rare autosomal dominant condition that affects about 1 in 50,000 people. HAE-GL p.6 More than 450 different mutations in SERPING1, the gene for C1-INH, cause it. HAE-GL p.6 In about 20–25% of patients, a de novo SERPING1 mutation causes the disease. HAE-GL p.6 A family history may be absent in up to 25% of patients. HAE-GL p.6 Each child of an affected parent has a 50% chance of inheriting HAE. HAE-GL p.11

4.3 Pathophysiology: bradykinin, not histamine

C1-INH is a serine protease inhibitor (serpin). It is the major inhibitor of C1r, C1s, MASP1 and MASP2, plasma kallikrein and factor XIIa. It is a minor inhibitor of plasmin. HAE-GL p.6 In complement terms (M5), C1-INH controls the classical and lectin pathways. Abbas p.311 It also inhibits kallikrein and factor XII, which both promote bradykinin formation. Abbas p.311

1. Factor XII autoactivates on contact with negatively charged surfaces. HAE-GL p.6
2. Factor XIIa converts prekallikrein to plasma kallikrein. C1-INH normally inhibits both enzymes. HAE-GL p.6
3. Plasma kallikrein cleaves high-molecular-weight kininogen (HMWK) and releases bradykinin, a nonapeptide. HAE-GL p.6
4. Bradykinin binds the bradykinin B2 receptor and increases vascular permeability. HAE-GL p.6
5. Metalloproteases, including ACE, rapidly degrade bradykinin. HAE-GL p.6

Bradykinin is the primary mediator of swelling in HAE-1/2. HAE-GL p.6 The missing brake also lets complement run: spontaneous activation produces excess cleaved C4 and C2. Janeway p.67 When C1-INH protein falls below 20–30% of normal, C4 and C2 breakdown increases. Abbas p.311 These changes fit the low C4 of HAE. HAE-GL p.6 Abbas also names a C2 fragment, "C2 kinin", as a mediator. Abbas p.311



D20. Bradykinin cascade with drug target pins (green = on-demand and prophylaxis; red = on-demand only). Licences follow the 2017 guideline. HAE-GL p.6 HAE-GL p.8 HAE-GL p.9–10

> **Why the anaphylaxis drugs fail**
>
> A failure to respond to antihistamines, glucocorticoids or epinephrine is one of the clues that raise suspicion of HAE. HAE-GL p.6 These drugs act against mast cell mediators. HAE swelling comes from bradykinin acting on the B2 receptor. HAE-GL p.6

4.4 Clinical features and triggers

Suspect HAE-1/2 in recurrent angioedema with: a positive family history, onset in childhood or adolescence, recurrent painful abdominal attacks, upper airway edema, no response to antihistamines, glucocorticoids or epinephrine, prodromal signs, and no wheals. HAE-GL p.6 Symptoms usually start in childhood or adolescence. The median age of onset is about 12 years. Subcutaneous edema is the most common and earliest symptom. HAE-GL p.10 Abbas describes edema of skin and mucosa with abdominal pain, vomiting, diarrhoea and possible airway obstruction. Abbas p.311

Erythema marginatum is a prodromal sign in 42–58% of paediatric cases. It is often mistaken for urticaria. HAE-GL p.11 Wheals do not exclude HAE, because CSU can coexist. The absence of wheals does not exclude mast cell-mediated angioedema. HAE-GL p.7

Triggers include trauma (dental, medical, surgical), oestrogen-containing contraceptives and hormone replacement, psychological stress, fatigue, febrile illness and menstruation. Most attacks are unpredictable. HAE-GL p.14 ACE inhibitors can increase or precipitate attacks; avoid them strictly. HAE-GL p.14 Procedures with mechanical impact on the upper aerodigestive tract (dental work, intubation, bronchoscopy, endoscopy) can trigger swelling, usually within 48 hours. HAE-GL p.8–9

> **Clinical correlation**
>
> Mast cell-mediated angioedema is far more common than HAE-1/2. When the diagnosis is not yet clear and the history does not fit HAE, give antihistamines and, if needed, epinephrine and corticosteroids. HAE-GL p.7

4.5 Diagnosis recommendations

- **Rec. 1**: test C1-INH function, C1-INH protein and C4 in all suspected patients; repeat low results. HAE-GL p.6
- **Rec. 13**: test children from HAE families as soon as possible, including all offspring of an affected parent. HAE-GL p.11
- **Rec. 20**: screen family members, because the first attack can affect the airway and be fatal. HAE-GL p.15

4.6 On-demand treatment

- **Rec. 2**: consider every attack for on-demand treatment. Treat any attack that affects, or may affect, the upper airway. HAE-GL p.7
- **Rec. 3**: treat attacks as early as possible. Early treatment shortens the time to resolution and the attack duration. HAE-GL p.7
- **Rec. 4**: treat attacks with C1-INH, ecallantide or icatibant. If none is available, use SDP; if SDP is not available, use FFP. HAE-GL p.7–8
- **Rec. 5**: consider intubation or a surgical airway early in progressive upper airway edema. Laryngeal attacks are emergencies. HAE-GL p.8
- **Rec. 6**: every patient has on-demand medication for two attacks and carries it at all times. HAE-GL p.8

Do not use tranexamic acid or androgens for attacks. They show no or minimal effect on-demand. HAE-GL p.8

Agent (2017)	Target	Key facts	Source
pdC1-INH (Berinert, Cinryze)	Replaces C1-INH	Half-life >30 h; negligible allergy risk; licensed for self-administration	HAE-GL p.7–8
rhC1-INH (Ruconest)	Replaces C1-INH	From transgenic rabbit milk; half-life ~3 h; adults and adolescents; contraindicated in rabbit allergy	HAE-GL p.8
Ecallantide	Plasma kallikrein	USA only; age 12+; anaphylaxis 3–4% → healthcare professional gives it	HAE-GL p.8
Icatibant	Bradykinin B2 receptor	Synthetic peptide antagonist; self-administered; adults (children or adolescents in some countries, 2017); half-life 1–2 h	HAE-GL p.8, p.11

4.7 Short-term (pre-procedural) prophylaxis

Rec. 7: give short-term prophylaxis before procedures that can induce an attack. Use C1-INH concentrate as close as possible to the start of the procedure; most experts give 1000 U or 20 U/kg pdC1-INH. HAE-GL p.9 Attenuated androgens (5 days before and 2–3 days after) are an older alternative. Most experts do not recommend tranexamic acid for this purpose. Breakthrough attacks can occur, so on-demand treatment must be available. HAE-GL p.9

4.8 Long-term prophylaxis

Rec. 8–9: consider prophylaxis before life events with high disease activity. Evaluate every patient for long-term prophylaxis at every visit. HAE-GL p.9

Line (2017)	Agent	Notes	Source
First line (Rec. 10)	C1-INH (pdC1-INH)	Twice weekly, IV or subcutaneous (40 or 60 U/kg)	HAE-GL p.9–10
Second line (Rec. 11)	Attenuated androgens (e.g., danazol)	Many side effects (virilization, menstrual disorders, weight gain, acne); semiannual blood/urine tests and yearly liver ultrasound; dose by clinical response, not by C4 or C1-INH level	HAE-GL p.10
Not recommended	Antifibrinolytics (tranexamic acid)	Used mainly when C1-INH is unavailable and androgens are contraindicated	HAE-GL p.10

4.9 Special populations

Children. Rec. 14: use C1-INH for attacks in children under 12. The guideline states that pdC1-INH is the only approved on-demand drug in childhood. HAE-GL p.11 It also notes that in 2017, rhC1-INH and ecallantide were licensed for adolescents, and icatibant for children or adolescents, in some countries. HAE-GL p.11 Do not use androgens for long-term prophylaxis before Tanner stage V. If C1-INH is unavailable, tranexamic acid is preferred. HAE-GL p.12 C4 is not useful below 12 months of age. Cord blood gives falsely low values. Repeat early tests after age 1. HAE-GL p.11

Pregnancy and lactation. C1-INH levels fall in normal pregnancy. Interpret tests with caution and repeat them after delivery. HAE-GL p.12 **Rec. 15**: C1-INH is the preferred therapy for attacks during pregnancy and lactation. Androgens are contraindicated in pregnancy because they cross the placenta. HAE-GL p.13 C1-INH concentrate is also first-line therapy in pregnant or breastfeeding patients. It is safe and effective when long-term prophylaxis is needed in pregnancy. HAE-GL p.12–13

4.10 Self-administration and action plan

- **Rec. 16**: every patient has an individual action plan and carries an HAE identification card. HAE-GL p.13
- **Rec. 18**: teach self-administration to every patient who has an on-demand drug licensed for it. IV self-administration is possible from age 8. HAE-GL p.14
- All C1-INH concentrates and icatibant are licensed for self-administration; indications vary by country. HAE-GL p.7
- With upper airway swelling: self-treat, then go to emergency care. Swelling can progress or rebound. HAE-GL p.14

5. PNH and aHUS: diseases of regulator defects

5.1 Paroxysmal nocturnal haemoglobinuria (PNH)

CD59 is GPI-linked. It blocks the addition of C9 to C5b-8 and so prevents MAC formation on host cells. Abbas p.312 Mutations in PIG-A cause PNH through defective GPI anchors for CD59 and DAF. Abbas p.315 The mutation is not inherited. It is an acquired somatic mutation in haematopoietic stem cells. Abbas p.312 Recurrent intravascular haemolysis causes chronic haemolytic anaemia and venous thrombosis. Abbas p.312

> **Source inconsistency**
>
> One summary sentence in Abbas (p.315) groups DAF deficiency with C1-INH deficiency as a cause of HAE. Elsewhere, Abbas and Janeway link loss of DAF and CD59 to PNH, not to angioedema. Abbas p.315 Do not use that sentence for HAE.

5.2 Atypical haemolytic uraemic syndrome (aHUS)

aHUS involves defective complement regulation. The most common mutations are in the factor H gene. The MCP gene is also mutated in many patients. Abbas p.315 Children present with microangiopathic haemolytic anaemia, thrombocytopenia and acute renal failure. Mutant factor H or MCP binds C3b and C4b on endothelium less well. Abbas p.315 Deficiencies in MCP, factor I or factor H can cause aHUS. Heterozygous mutations are enough to predispose. Janeway p.70 Anti-C5 antibodies are approved for PNH and aHUS. Abbas p.316 Detailed treatment of PNH and aHUS is not covered in provided sources — needs expert input.

5 key points

1. Early classical deficiency (C1q, C4, C2) → SLE-like immune-complex disease; C2 deficiency is the most common. Abbas p.315
2. C3 deficiency, or C3 depletion from factor I deficiency → severe pyogenic infections. Abbas p.315
3. C5–C9, properdin (X-linked) and factor D/B deficiency → *Neisseria* infections. Abbas p.315
4. HAE-1/2 is autosomal dominant (SERPING1) and bradykinin-mediated. Diagnose with C1-INH function, C1-INH protein and C4; use C1q to find acquired deficiency. HAE-GL p.6–7
5. 2017 guideline: treat attacks early with C1-INH, ecallantide or icatibant; C1-INH is first-line LTP; androgens are second-line LTP. HAE-GL p.7–10

3 exam traps

1. Normal C1-INH level and function do not exclude HAE. Think of HAE nC1-INH ("type III") in a patient with a family history. HAE-GL p.6
2. Adrenaline, antihistamines, glucocorticoids, tranexamic acid and androgens do not treat an acute HAE attack. HAE-GL p.6–8
3. PNH is not an inherited complement deficiency: the PIG-A mutation is somatic. Abbas p.312 DAF and CD59 loss causes haemolysis, not angioedema. Abbas p.315

Practice: M6 questions in the quiz (IDs M6-A1 … M6-K3).

Sources: Abbas, Cellular and Molecular Immunology 11th ed., pp. 311, 312, 315, 316; Janeway's Immunobiology 8th ed., pp. 59, 67, 70, 532, 533; WAO/EAACI HAE guideline 2017 revision (Allergy 2018), pp. 5–15 · Fact IDs refer to `ledger/facts.csv`.